Points to Remember

- Acid is a substance which furnishes H^+ ions or H_3O^+ ions when dissolved in water.
- Base is a substance which releases OH^- ions when dissolved in water.
- Salt is the product of reaction between acids and bases.
- Acids and bases neutralize each other to form corresponding salts and water.
- Salts have various uses in everyday life.
- Acidic and basic solutions in water conduct electricity because they produce hydrogen and hydroxide ions respectively.
- When an acid reacts with a metal, hydrogen gas is evolved and a corresponding salt is formed.
- Phenolphthalein and Methyl orange are used as indicators to find out whether the given solution is acid or base.
- Litmus paper is also used to find out whether the given solution is acid or base.
- pH paper is find out the given solution whether acidic or basic in nature.
- Aquaregia is a mixture of hydrochloric acid and nitric acid optimally in a molar ratio of 3:1
- pH Scale is used to find out the power of hydrogen ion concentration in a solution.

A-Z GLOSSARY

Acids	Substance which furnishes H^+ ions H_3O^+ ions when dissolved in water.
Bases	Substance which furnishes OH^- ions when dissolved in water.
Salts	Product of reaction between acids and bases.
Indicators	Chemical substances used to find out whether the given solution is acid or base.
pH Scale	Scale used to find out Hydrogen ion concentration in a solution.
pH Paper	Paper used to find out whether the given solution is acidic or basic or neutral in nature.
Aquaregia	Mixture of hydrochloric acid and nitric acid prepared optimally in a molar ratio of 3 : 1.
Hygroscopic substance	Substance which absorbs water from the surroundings.

46GNM

TEXTBOOK EXERCISES

I. Choose the correct answer.

1. $Zn + 2\ HCl \rightarrow ZnCl_2 + \ldots\uparrow (H_2, O_2, CO_2)$
2. Apple contains malic acid. Orange contains _____ (oxalic acid, ascorbic acid).
3. Acids in plants and animals are organic acids. Whereas Acids in rocks and minerals are _____ (Inorganic acids, Weak acids).
4. Acids turn blue litmus paper to _____ (green, red, orange).
5. Since metal carbonate and metal bicarbonate are basic, they react with acids to give salt and water with the liberation of _____ (NO_2, SO_2, CO_2).
6. The hydrated salt of copper sulphate has _____ colour (red, white, blue).

II. Answer briefly.

1. Classify the various types of Acids based on their sources.
2. Write any four uses of acids.
3. Give the significance of pH of soil in agriculture.
4. What are the various uses of Aquaregia.
5. What are the uses of Plaster of Paris?
6. Two acids 'A' and 'B' are given. Acid A gives one hydrogen ion per molecule of the acid in solution. Acid B gives two hydrogen ions per molecule of the acid in solution.

IX_SCI_EM_Unit-14_CHE.indd 170 12/7/2022 4:28:09 PM